

Pre-High Engine Temperature Lamp Driver Malfunction

Symptoms

- The pre-high engine temperature warning lamp is illuminated when a pre-high engine temperature condition does **not** exist.
- The pre-high engine temperature warning lamp is **not** illuminated when a pre-high engine temperature condition exists.

How To Use This Tree

This symptom tree can be used to troubleshoot a pre-high temperature lamp driver malfunction. Start by performing Step 1 troubleshooting. Step 2 will ask a series of questions and will provide a list of troubleshooting steps to perform, depending on the symptom.

Shoptalk

Troubleshooting Summary

STEPS		SPECIFICATIONS
STEP 1.	Check Customer Interface Box Wiring	
	STEP 1A. Check Pre-High Engine Temperature Warning Signal Wire for Open	Less than 10 ohms?
	STEP 1B. Check Pre-High Engine Temperature Warning Signal Wire for Wire to Wire Short	Less than 10 ohms?
	STEP 1C. Check Pre-High Engine Temperature Warning Signal Wire for Short to Ground	Less than 10 ohms?
STEP 2.	Check Engine Harness to Customer Interface Box Cable	
	STEP 2A. Check Pre-High Engine Temperature Warning Signal Wire for Open	Less than 10 ohms?

STEPS		SPECIFICATIONS
	STEP 2B. Check Pre-High Engine Temperature Warning Signal Wire for Wire to Wire Short	Less than 10 ohms?

STEP 1. Check Customer Interface Box Wiring

STEP 1A. Check Pre-High Engine Temperature Warning Signal Wire for Open

Conditions :

- Open the customer interface box
- Disconnect customer interface box to engine harness cable connector C2 from the customer interface box.

Action	Specification/Repair	Next Step
Check the pre-high engine temperature warning signal wire for an open. • Place one test lead on the pre-high engine temperature warning signal pin in connector C2. • Place the other test lead on pre-high engine temperature warning signal terminal on the X4 connector.	Less than 10 ohms? YES	1B
	Less than 10 ohms? NO Repair : Replace the wire. Refer to Procedure 015-023 (/qs3/pubsys2/xml/en/procedures/116/116-015-023.html).	Repair complete

STEP 1B. Check Pre-High Engine Temperature Warning Signal Wire for Wire to Wire Short

Conditions :

- Open the customer interface box
- Disconnect customer interface box to engine harness cable connector C2 from the customer interface box.

Action	Specification/Repair	Next Step
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Check the pre-high engine temperature warning signal wire for wire to wire short. - Place one test lead on the pre-high engine temperature warning signal pin in connector C2. - Place the other test lead on each of the remaining terminals in X4 connector.	Less than 10 ohms? YES Repair : Replace the wire. Refer to Procedure 015-023 (/qs3/pubsys2/xml/en/procedures/116/116-015-023.html).	Repair complete
	Less than 10 ohms? NO	1C

STEP 1C. Check Pre-High Engine Temperature Warning Signal Wire for Short to Ground

Conditions :

- Open the customer interface box
- Disconnect customer interface box to engine harness cable connector C2 from the customer interface box.

Action	Specification/Repair	Next Step
Check the pre-high engine temperature warning signal wire for short to ground. - Place one test lead on the pre-high engine temperature warning signal pin in connector C2. - Place the other test lead on panel ground.	Less than 10 ohms? YES Repair : Replace the wire. Refer to Procedure 015-023 (/qs3/pubsys2/xml/en/procedures/116/116-015-023.html).	Repair complete
	Less than 10 ohms? NO	2A

STEP 2. Check Engine Harness to Customer Interface Box Cable

STEP 2A. Check Pre-High Engine Temperature Warning Signal Wire for Open

Conditions :

- Disconnect customer interface box to engine harness cable connector C2 from the customer interface box
- Disconnect customer interface box to engine harness cable connector C9 from the engine harness.

Action	Specification/Repair	Next Step
Check pre-high engine temperature warning signal wire for an open. • Place a jumper between the pre-high engine temperature warning signal pin and the high engine temperature (HET) shutdown signal pin in the C9 connector. • Place one test lead in the pre-high engine temperature warning signal pin of the C2 connector. • Place the other test lead in the high engine temperature (HET) shutdown signal pin of the C2 connector.	Less than 10 ohms? YES	2B
	Less than 10 ohms? NO Repair : Replace the cable.	Repair complete

STEP 2B. Check Pre-High Engine Temperature Warning Signal Wire for Wire to Wire Short**Conditions :**

- Disconnect customer interface box to engine harness cable connector C2 from the customer interface box
- Disconnect customer interface box to engine harness cable connector C9 from the engine harness.

Action	Specification/Repair	Next Step

Check pre-high engine temperature warning signal wire for wire to wire short. - Place one test lead on the pre-high engine temperature warning signal pin in connector C2. - Place the other test lead on each of the remaining pins in the C2 connector.	Less than 10 ohms? YES Repair : Refer to Section TF in the Troubleshooting and Repair Manual, Electronic Control System, QSK19 CM850 Modular Common Rail System Series Engines, Bulletin 4021493 (/qs3/portal/manualviewer.html?path=/qs3/pubsys2/xml/en/outlines/4021493.html), or the Troubleshooting and Repair Manual, Electronic Control System, QSK38, QSK50, and QSK60, CM850 Modular Common Rail System Series Engines, Bulletin 4021533 (/qs3/portal/manualviewer.html?path=/qs3/pubsys2/xml/en/outlines/4021533.html) or refer to the OEM service manual.	Repair complete
	Less than 10 ohms? NO Repair : Replace the cable.	Repair complete